

IMPACT OF WEATHER VARIABLES ON SHORT-TERM ELECTRICITY DEMAND FORECASTING IN SINGAPORE: A COMPARATIVE STUDY WITH PPSO-OPTIMISED MODELS



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Declaration of Contributions

We confirm that the work contained in this group project report has been composed solely by the members of our group and has not been accepted in any previous application for a degree. All sources of information have been specifically acknowledged and all verbatim extracts are distinguished by quotation marks.

The contribution of each member is declared as follows:

Member	Contribution	Share
.....	Data acquisition & cleaning (EMA demand parsing, Open-Meteo integration, data validation tools)	16.7%
.....	Feature engineering & EDA (CCI index, lag/rolling features, exploratory visualisations)	16.7%
.....	Model development (baseline, Linear Regression, Random Forest, XGBoost, CatBoost)	16.7%
.....	PPSO optimiser implementation & fair-tuning experiment design	16.7%
.....	Web portal (FastAPI backend, React frontend, deployment)	16.7%
.....	Evaluation, analysis & report writing (metrics, figures, LaTeX)	16.7%

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1 Introduction

In Singapore, a compact grid serves approximately 5.9 million residents under a tropical climate that sustains year-round cooling demand; national electricity consumption exceeded 55 TWh in 2024 (Energy Market Authority of Singapore 2025). Because air-conditioning is the largest end-use of electricity in buildings, demand is highly sensitive to small variations in temperature and humidity (Zhao and Magoulès 2012), with holidays, weekends, and monsoon seasons adding further variability. Accurate short-term forecasting is therefore essential for reliable grid operation and cost-effective generation scheduling.

Classical methods (ARIMA, exponential smoothing, regression) assume linear, stationary relationships and struggle with the non-linear interactions among weather, calendar, and autoregressive demand dynamics (Al-Musaylh et al. 2018). Tree-based machine learning models capture these interactions well but are sensitive to hyperparameters, and manual tuning is expensive and often sub-optimal. Zhang et al. (Zhang, Y. Chen, and Yan 2023) showed that CatBoost paired with Phasor Particle Swarm Optimisation (PPSO) outperforms five other meta-heuristics—but on hourly data from a *single residential customer*, and with tuning applied only to the favoured model. Whether the approach generalises to *daily national-level* demand in a tropical setting, and whether its advantage survives when *all* competing models receive the same tuning budget, remain open questions.

This project (i) builds a daily Singapore dataset merging EMA demand with Open-Meteo weather data (February 2023 – December 2025); (ii) engineers 16 features across weather, derived-comfort, calendar, and autoregressive categories; (iii) compares eight model configurations—a Lag-1 baseline, Linear Regression, Random Forest, XGBoost, and CatBoost, plus PPSO-tuned variants of each tree model under an *identical* budget—on six metrics; and (iv) deploys the best model through a web portal (FastAPI + React) for non-technical users. Intra-day resolution, sector decomposition, and deep learning models are out of scope.

2 Related Work

Classical statistical methods (ARIMA, Holt-Winters) capture linear temporal dependencies but are consistently outperformed by machine learning when exogenous weather predictors matter: SVR and MARS beat ARIMA on short-term Australian demand (Al-Musaylh et al. 2018), and tree-based methods dominate linear regression for residential consumption (Schirmer, Mporas, and Potamitis 2019). Among tree ensembles, Random Forest (Breiman 2001) reduces variance through bagging; XGBoost (T. Chen and Guestrin 2016) adds regularised sequential boosting and is widely used in energy applications (Khan, Byun, et al. 2020); CatBoost (Prokhorenkova et al. 2018) contributes ordered boosting and native categorical-feature handling, which suits demand data rich in calendar categories. Weather, holidays, and weekends are recognised as key influencing factors (Khan, Kim, et al. 2021).

Meta-heuristic hyperparameter tuning has shown large gains—e.g. 42% accuracy improvement from PSO-optimised models on university buildings (Li et al. 2020). PPSO (Ghasemi et al. 2019) reformulates PSO with a phase angle that modulates cognitive and social terms through periodic functions, eliminating manual tuning of inertia and acceleration coefficients. Zhang et al. (Zhang, Y. Chen, and Yan 2023) combined CatBoost with PPSO for hourly consumption of a single customer and found it superior to five alternative optimisers.

Gap. Prior work tunes only a *single* favoured model, leaving open whether reported gains reflect the algorithm or preferential tuning, and the CatBoost-PPSO methodology has not been evaluated on daily national-level demand in a tropical climate. This project addresses both gaps.

3 Methodology

3.1 Pipeline Overview

The system is a modular, end-to-end pipeline (Figure 1) driven by a single YAML configuration: data acquisition, feature engineering, chronological train/test split, model training with PPSO tuning, and multi-metric evaluation. Each stage is an independent Python module.

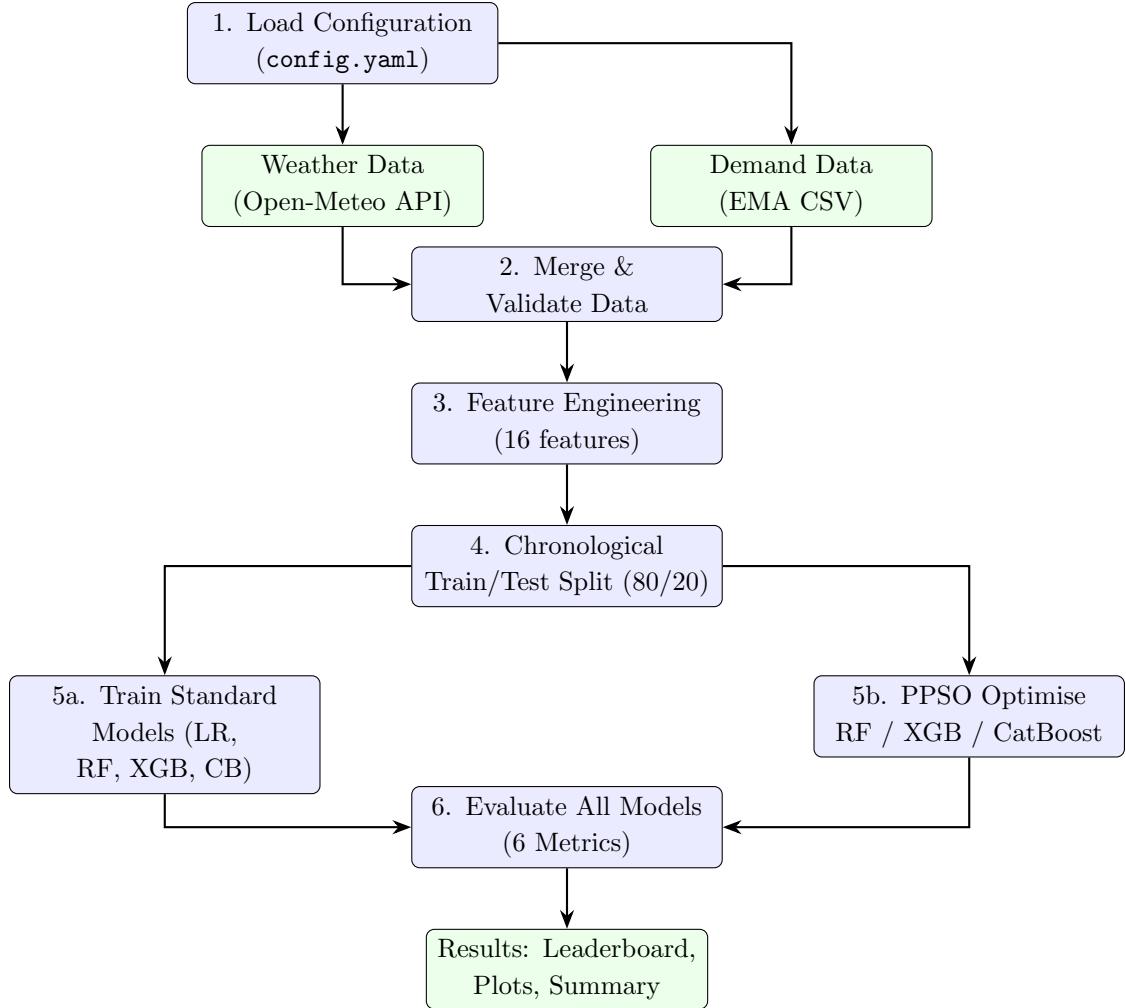


Figure 1: High-level architecture of the electricity demand forecasting pipeline.

3.2 Data and Feature Engineering

Daily national demand (MWh) is obtained from EMA (Energy Market Authority of Singapore 2025); four daily weather variables (mean temperature, mean relative humidity, maximum wind speed, cumulative solar radiation) are retrieved from the Open-Meteo Archive API (Open-Meteo 2025) for Singapore’s coordinates (1.3521°N, 103.8198°E). An inner join on date yields 1,065 observations (February 2023 – December 2025). Figure 2 shows the demand and weather series over the three-year period: demand exhibits clear weekly cyclicality and a mild upward trend, while temperature oscillates within the narrow 26–30°C tropical band.

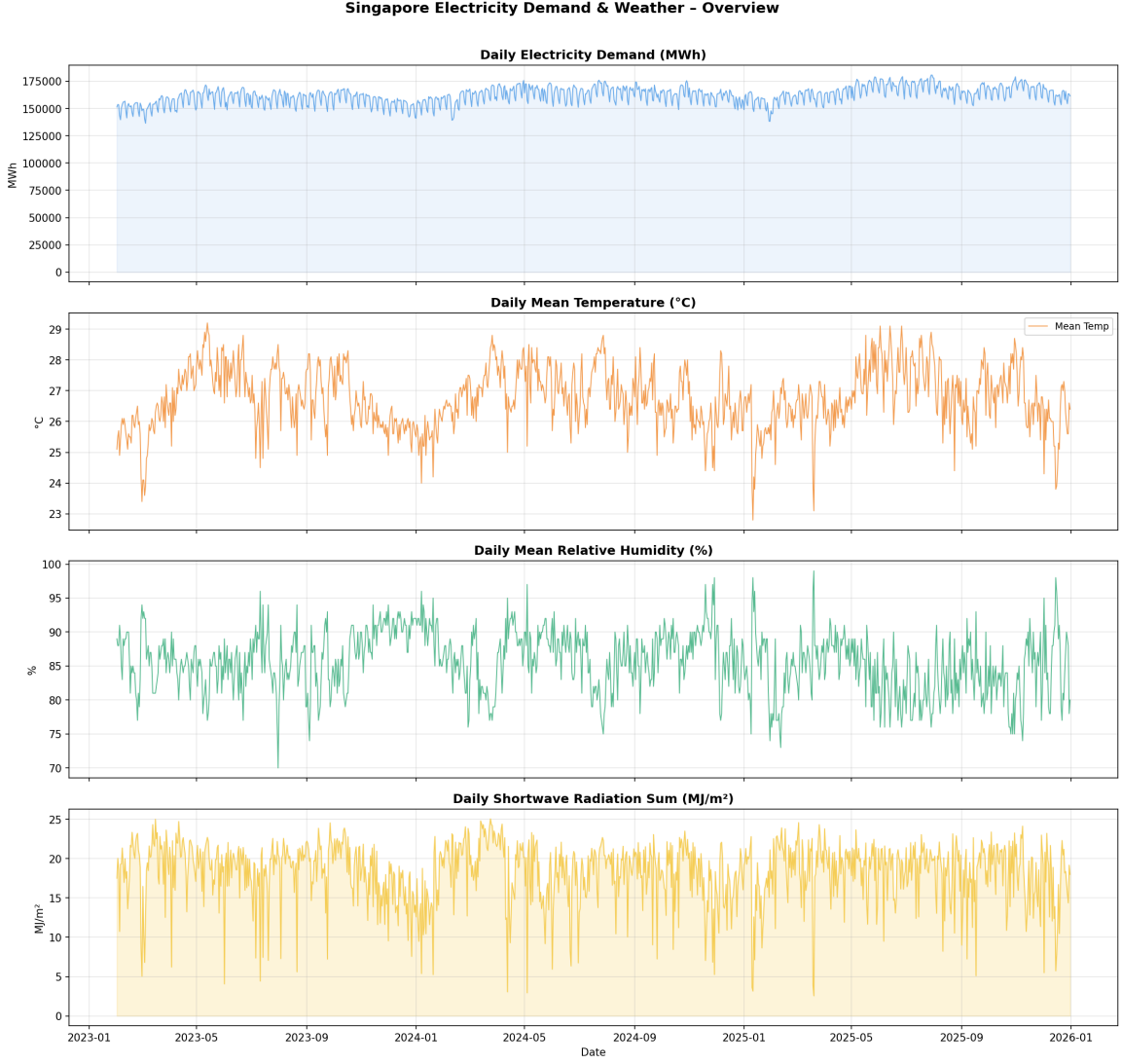


Figure 2: Daily electricity demand and weather variables in Singapore, February 2023 – December 2025.

Sixteen features are constructed across four categories: *weather* (the four raw variables); *derived* — a Comfort Climate Index, $CCI = T \times (1 + 0.01 \times H)$, capturing combined thermal stress, plus a season indicator; *temporal* (day-of-week, month, weekend and Singapore public-holiday flags); and *autoregressive* (demand lags at 1 and 7 days; 3- and 7-day rolling means and standard deviations). An 80/20 *chronological* split (852 train / 213 test samples) prevents forward-looking bias; features are standardised only for Linear Regression.

3.3 Models and Fair Tuning Protocol

Eight configurations are compared: a Lag-1 baseline, Linear Regression, and default Random Forest, XGBoost, and CatBoost, plus PPSO-tuned variants **RF-PPSO**, **XGB-PPSO**, and **CatBoost-PPSO**. The central methodological decision is to apply PPSO to *every* tunable model under an *identical* budget, rather than only to CatBoost as in the reference study (Zhang, Y. Chen, and Yan 2023). This isolates each algorithm’s intrinsic merit from the benefit of tuning.

PPSO was selected for three pragmatic reasons: (i) it is *parameter-free*—the phasor angle adaptively balances exploration and exploitation, avoiding meta-tuning of the optimiser itself, which would re-introduce experimenter bias; (ii) its periodic velocity modulation suits the non-convex, expensive

hyperparameter landscape where each fitness call requires several full model trainings; and (iii) it was validated against six meta-heuristics in the closest prior study (Zhang, Y. Chen, and Yan 2023). Mathematical details follow Ghasemi et al. (Ghasemi et al. 2019) and are not repeated here.

Each model’s four-dimensional search space covers its most sensitive hyperparameters (Table 1). The shared budget is 15 particles $\times$ 20 iterations with early stopping after 5 stagnant iterations; fitness is the mean RMSE over 3-fold `TimeSeriesSplit` cross-validation on the training set only (Figure 3). All seeds are fixed to 42.

Table 1: PPSO hyperparameter search spaces (identical budget for all models).

Model	Tuned hyperparameters (bounds)			
Random Forest	trees (100–800)	depth (4–25)	min split (2–20)	min leaf (1–10)
XGBoost	trees (100–1000)	depth (3–10)	learn. rate (0.01–0.2)	L_2 (1–20)
CatBoost	iterations (100–800)	depth (4–10)	learn. rate (0.01–0.2)	L_2 (1–20)

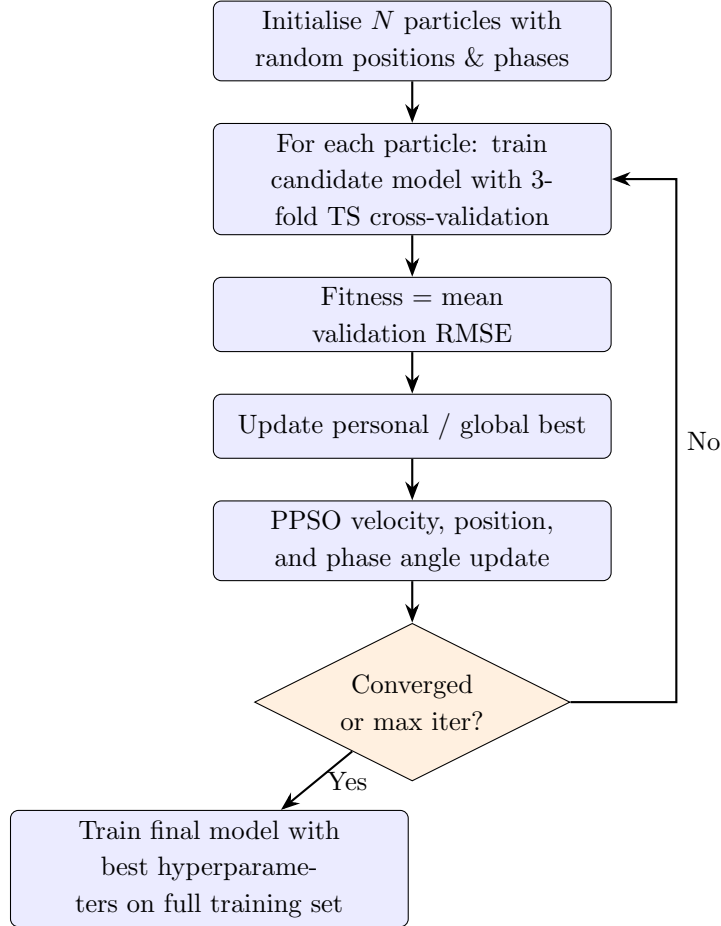


Figure 3: PPSO hyperparameter optimisation workflow (applied identically to RF, XGBoost, and CatBoost).

3.4 Evaluation Metrics

Following Zhang et al. (Zhang, Y. Chen, and Yan 2023), six metrics are computed on the held-out test set: MAE, RMSE, MAPE, and RAE (errors; lower is better) plus R^2 and the Willmott Index (WI) (Willmott 1982) (agreement; higher is better). WI is included for its robustness to outliers relative to R^2 .

3.5 Web Portal

The best model is deployed via a lightweight portal: a FastAPI backend loads the serialised model and exposes REST endpoints (`/api/summary`, `/api/range`, `/api/predict`, `/api/forecast`), and a React/Vite frontend offers a *History Report* page (retrospective accuracy over a chosen period, Figure 4) and a *Demand Trends* page (forward outlooks driven by live Open-Meteo weather forecasts).

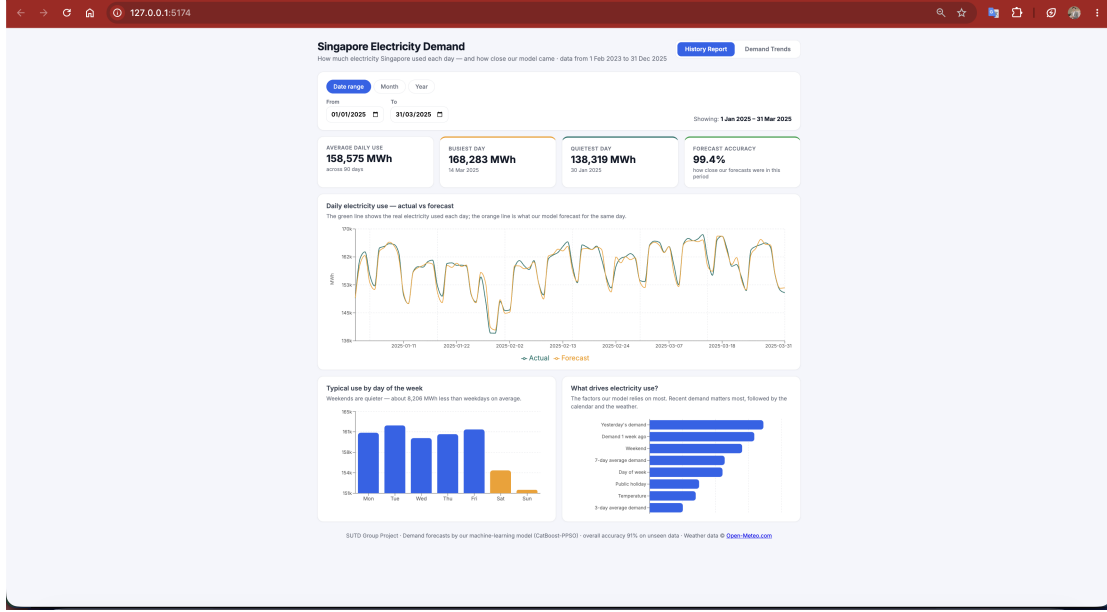


Figure 4: History Report page of the web portal: actual vs. predicted demand and summary statistics for a user-selected period.

4 Results

4.1 Model Comparison

Table 2 reports test-set performance for all configurations, sorted by RMSE.

Table 2: Model comparison on the test set (213 samples), sorted by RMSE. Best values in **bold**.

Model	MAE	RMSE	R^2	MAPE (%)	RAE	WI
CatBoost-PPSO	1,496.2	1,895.5	0.9135	0.89	0.280	0.9774
CatBoost	1,679.4	2,118.8	0.8919	0.99	0.314	0.9709
Linear Reg.	1,693.4	2,119.2	0.8919	1.01	0.317	0.9731
XGB-PPSO	1,687.8	2,165.6	0.8871	1.00	0.315	0.9697
Random Forest	1,774.0	2,237.6	0.8795	1.05	0.332	0.9653
RF-PPSO	1,794.1	2,283.9	0.8744	1.06	0.335	0.9634
XGBoost	1,833.4	2,313.4	0.8712	1.08	0.343	0.9652

Key observations:

1. **CatBoost-PPSO leads on all six metrics:** $R^2 = 0.914$ (over 91% of demand variance explained), RMSE of 1,895.5 MWh ($\approx 1.1\%$ of mean daily demand), MAPE of 0.89%.
2. **PPSO benefits are model-dependent.** PPSO improves CatBoost (RMSE -10.5%) and

XGBoost (-6.4%), but slightly degrades Random Forest ($+2.1\%$), indicating mild overfitting of the cross-validated fitness to the training period. Because all models received the same budget, these deltas make clear that CatBoost-PPSO’s lead is not an artefact of preferential tuning.

3. **Linear Regression is competitive**, tied with default CatBoost on RMSE—evidence that the engineered autoregressive features (lag-1, rolling means) render much of the problem linear.
4. **All models exceed $R^2 = 0.87$ and $WI = 0.96$** , confirming the 16-feature set captures the dominant demand drivers.

4.2 Prediction Behaviour

Figure 5 shows actual versus predicted demand for CatBoost-PPSO over the 213-day test period. The model reproduces weekly cyclicality (weekend dips), tracks demand level shifts, and errs most on extreme holiday-related drops.

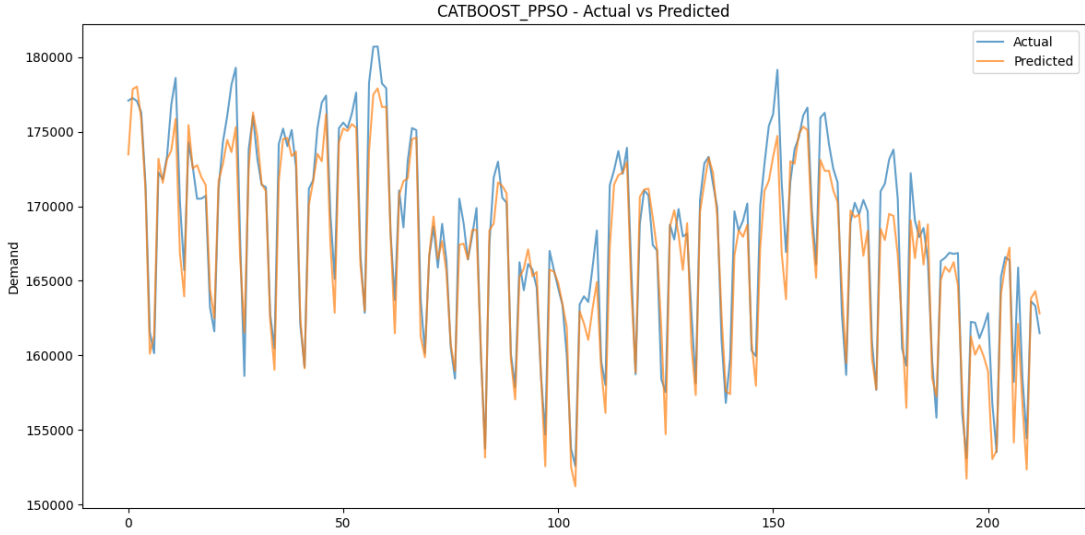


Figure 5: Actual vs. predicted daily electricity demand (MWh), CatBoost-PPSO, test set.

4.3 Feature Importance

Figure 6 presents CatBoost-PPSO importance scores. Autoregressive features dominate (`demand_lag_1`, rolling means), followed by `day_of_week`; weather variables (temperature, humidity, CCI) contribute meaningful mid-rank exogenous signal. The same hierarchy appears across RF and XGBoost, confirming robustness.

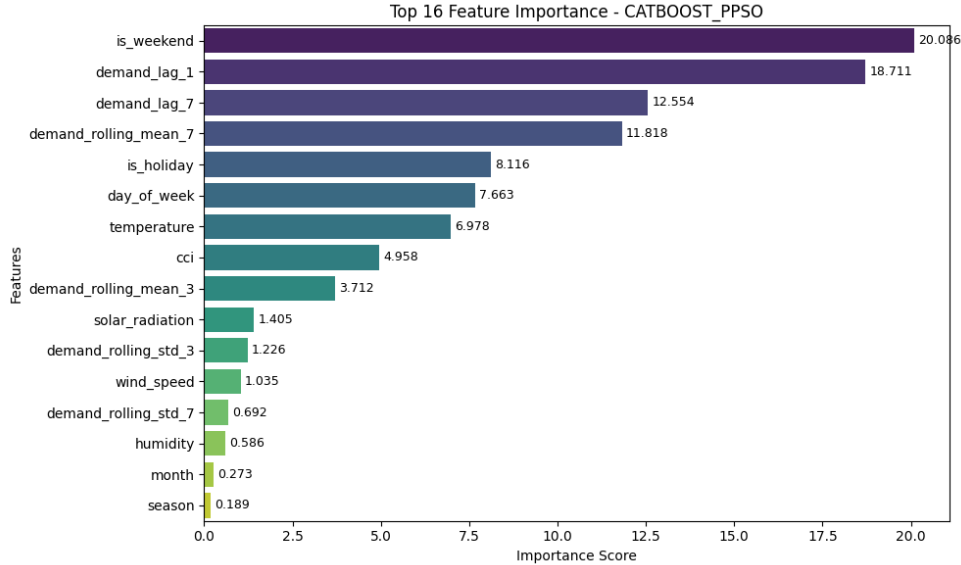


Figure 6: Feature importance, CatBoost-PPSO. Autoregressive features dominate, with weather variables providing secondary signal.

4.4 Impact of PPSO Optimisation

Table 3 quantifies the tuning effect on CatBoost, the model with the largest gain.

Table 3: Impact of PPSO hyperparameter optimisation on CatBoost.

Metric	CatBoost (default)	CatBoost-PPSO	Improvement
MAE (MWh)	1,679.4	1,496.2	−10.9%
RMSE (MWh)	2,118.8	1,895.5	−10.5%
R^2	0.8919	0.9135	+2.2 pp
MAPE (%)	0.99	0.89	−10.6%
RAE	0.314	0.280	−10.9%
WI	0.971	0.977	+0.6 pp

Compared with the reference study (Zhang, Y. Chen, and Yan 2023) ($R^2 = 0.971$ on hourly single-customer data), the lower R^2 here (0.914) is expected: daily national demand aggregates heterogeneous consumption across millions of consumers, adding unexplained variance. The methodology nonetheless transfers effectively across scale, resolution, and climate.

5 Discussion

Interpretation. Three conclusions emerge beyond raw rankings. First, hyperparameter optimisation is *not* universally beneficial: PPSO helped the gradient-boosted models, whose performance hinges on learning rate and regularisation, but not Random Forest, whose bagging mechanism is robust to hyperparameters and where tuning instead overfitted the cross-validation folds. Reporting only a tuned flagship model—as is common in the literature—would have concealed this. Second, the strength of Linear Regression shows that autoregressive features carry most of the daily-resolution signal; the value of non-linear models lies in weather–calendar interactions, precisely where CatBoost’s categorical handling helps. Third, time-series cross-validation inside the PPSO fitness function was essential: random K-fold produced optimistic fitness values that did not

transfer to the test period.

Limitations. Daily aggregation masks intra-day peaks; Open-Meteo reanalysis data may deviate from station measurements; external shocks (economic events, policy changes) are not modelled; PPSO is stochastic and results reflect a single seeded run; multi-day portal forecasts inherit upstream weather-forecast error.

Future work. Hourly resolution with intra-day features; multi-seed PPSO runs with significance testing; sequence models (LSTM, Transformer) under the same fair-tuning protocol; online retraining in the deployed portal.

6 Conclusion

This project developed a complete pipeline for daily electricity demand forecasting in Singapore, comparing eight model configurations on a 16-feature dataset of weather, calendar, and autoregressive variables (February 2023 – December 2025). The key takeaways are:

1. **CatBoost-PPSO is the best model**, achieving $R^2 = 0.914$, MAE of 1,496.2 MWh, RMSE of 1,895.5 MWh, and MAPE of 0.89% on the held-out test set.
2. **Fair tuning changes the story**: applying an identical PPSO budget to all tree models showed the gain is substantial for CatBoost (-10.5% RMSE), moderate for XGBoost (-6.4%), and absent for Random Forest—so the winning margin reflects the algorithm, not preferential optimisation.
3. **Autoregressive features dominate**; weather variables (temperature, humidity, CCI) provide meaningful secondary signal, consistent with a cooling-driven tropical demand profile.
4. **The methodology generalises**: the CatBoost-PPSO approach, originally validated on hourly single-customer data (Zhang, Y. Chen, and Yan 2023), transfers to daily national-level demand in an equatorial climate.
5. **Practical delivery**: the best model is operationalised in a web portal offering retrospective accuracy reports and weather-driven demand outlooks, demonstrating a path from research pipeline to decision-support tool for grid operators and energy planners.

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